

AKHILESH BABU TUMATI

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EDUCATION

Virginia Tech

Master of Engineering in Computer Science (GPA: 4.0 / 4.0)

Blacksburg, VA

Aug 2024 – May 2026

Vellore Institute of Technology

Bachelor of Technology in Computer Science (GPA: 3.5 / 4.0)

Vellore, India

Jul 2019 – Aug 2023

TECHNICAL SKILLS

Cloud Infrastructure (AWS & Azure): AWS Lambda, API Gateway, S3, CloudFront, CloudFormation (IaC), AWS CDK, IAM Roles & Policies, VPC, Security Groups, Route 53, CloudWatch Metrics/Alarms, Azure Cloud Services

DevOps, CI/CD & GitOps: GitLab CI/CD, Jenkins Pipelines, Docker Containerization, Kubernetes (EKS), Red Hat OpenShift, Automated Promotion Gates, Environment Provisioning, Release Engineering

Scripting & Programming Languages: Python, Bash, PowerShell, Java, TypeScript, JavaScript, SQL, C#/.NET Architecture Concepts

Databases & Storage: PostgreSQL (AWS RDS), Supabase, MongoDB Atlas, Couchbase, Database CI/CD & Migrations, Redis

Quality, Security & Monitoring: AWS CloudWatch, AWS X-Ray Concepts, Coverity Static Analysis, Black Duck Vulnerability Scanning, Jest, Supertest, Selenium, Katalon, TDD/BDD, High Availability (HA) & Disaster Recovery (DR)

Distributed Systems & AI: Microservices Architecture, Apache Kafka, Red Hat AMQ Streams, MirrorMaker, RAG Systems, LangChain

PROFESSIONAL EXPERIENCE

Virginia Tech

Cloud / Full Stack Developer (Part-time)

Blacksburg, VA

Jun 2025 – Dec 2025

- Engineered automated Infrastructure-as-Code (IaC) templates using CloudFormation and Serverless Framework to provision decoupled serverless microservices on AWS Lambda and API Gateway, cutting infrastructure costs by 25% versus EC2.
- Implemented resilient web tier hosting via Amazon S3 and CloudFront CDN distribution, configuring custom SSL/TLS certificates, CORS controls, and security headers with Helmet middleware.
- Hardened AWS IAM security posture by establishing least-privilege execution roles, VPC routing policies, AWS SES MFA authentication, and JWT revocation verification across all production API Gateway endpoints.
- Configured centralized operational monitoring and automated alerting using AWS CloudWatch metrics and alarms, capturing API latency deviations, error rates, and invocation limits.

UPS

Software Engineer (DevOps & Infrastructure)

Baltimore, MD

Aug 2023 – Aug 2024

- Enabled 20+ monthly zero-downtime releases and lowered deployment error rates by 30% by codifying declarative CI/CD pipelines, container promotion gates, and rollback automation in Jenkins for Kubernetes and OpenShift.
- Decreased query latency by 35% across a high-throughput Java and Couchbase distributed lookup service serving 10M+ daily operations by profiling execution plans, tuning memory allocations, and redesigning index topology.
- Architected cross-region disaster-recovery data synchronization pipelines across APAC, EU, and US clusters utilizing Apache Kafka, Red Hat AMQ Streams, and MirrorMaker, backed by automated Azure snapshots.
- Collaborated with back-end engineering teams to troubleshoot containerized service bottlenecks, monitor cluster resource utilization, and automate post-deployment health checks.

Honeywell

Software Developer Intern

Hyderabad, India

Jan 2023 – Jul 2023

- Automated CI/CD security and compliance gating within deployment pipelines by embedding Coverity static code analysis and Black Duck open-source vulnerability merge checks.
- Shortened manual QA release validation time by 40% and achieved 85% automated regression coverage across an enterprise aerospace platform serving 500+ clients by authoring 200+ automated test suites with Java and Selenium.
- Built reusable UI modules in Angular interfacing with Java Spring Boot REST microservices and JPA/Hibernate relational database models.

PROJECTS

MCCS Real-Time Transaction Risk Platform | Python, AWS/Cloud, MongoDB Atlas, LangChain, React

Sep 2024 – May 2025

- Deployed high-throughput Python risk analysis microservices delivering sub-second decision latencies on 15,000+ daily financial transactions.
- Automated 150+ monthly compliance audit packages and saved 12 hours/week of manual engineering preparation by developing an automated Gemini RAG pipeline using LangChain, FAISS, and MongoDB Atlas Vector Search.
- Built telemetry monitoring dashboards with React and Node.js to track operational model drift, latency spikes, and system anomalies in real time.

Nazeer and Nazeers Cloud E-Commerce Platform | PostgreSQL, Supabase, Next.js, TypeScript

Full-Stack / DevOps

- Implemented database CI/CD and automated migration tracking across 23 versioned SQL migrations using PostgreSQL and Supabase, enforcing schema integrity, immutable ledgers, and Row-Level Security (RLS) policies.
- Automated containerized web deployments and integrated transactional email alerts with real-time inventory reservation engines.

Evol Jewels Computer Vision Kiosk | *Docker, Next.js, PostgreSQL, Prisma, fal.ai*

Hackathon (2nd Place)

- Configured an interactive edge-device retail application with Next.js and Prisma ORM, integrating GPU-accelerated cloud inference APIs (fal.ai) with automated fallback pipelines.

CERTIFICATIONS & PUBLICATIONS

Certification: Nebius AI CloudOps Engineer Certification | Nebius (Issued Jul 2026 – Expires Jul 2029)

Publication: Tumati, A. B., “Face Invariant Classification and Detection of Mythological Characters (ClaDeMuC-CD).” *IEEE ICAECT 2023*.

Publication: Tumati, A. B., “Hybrid ML and DL models for flood level prediction.” *International Journal of Health Sciences*.